Pimpri Chinchwad Education Trust's
S.B.Patil College Of Science and Commerce
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UDISE NO. 27252001412
College Index No J 11 16 060

ACA/DI/21	Prelim 1 Examination	Academic Year :-2021-22
Rev:00		Total Marks:- 50
Class:- 12th Science		Date:- 30/12/2021
Time:- 9:00am – 12:00pm	Subject :- Computer Science 1	

Instructions

1. All questions are compulsory.
2. Assume suitable data wherever necessary.

Q. 1(A) Select the correct alternative and rewrite the following: [4]

(1) To place the image into a HTML file ______ attribute is used in IMG tag.
(a) <URL> (b) <ALT> (c) <SRC> (d)<HREF>

(2) Maximum number of nodes in a symmetric binary tree with depth four are______.
(a) 4 (b) 15 (c) 16 (d) 5

(3) Following is not a process state ______.
(a) ready (b) blocked (c) resumed (d) running

(4) ______is the name of the Web Browser.
(a) Embedded system (b) Netscape Navigator (c) Oracle (d) C++

(B) Answer any two of the following: [6]

1) Explain any 6 basic GUI features.
2) Explain the following data structures with suitable diagram:
i)Linear array ii)Linked List iii)Tree
3) Explain the general structure of HTML page.

Q.2(A) Answer any two of the following: [6]

1) Write a short on basic data types with their byte sizes.
2) What is searching? Explain Binary Search Algorithm.
3) Explain memory map of single user operating system.

(B) Answer any one of the following: [4]

1) Define Operating System .In which categories operating system provide its services?
2) Explain any four features of Windows NT Operating system.

Q.3(A) Answer any two of the following: [6]

1) Explain Internal and External fragmentation in memory management of operating system.
2) List any six data structure operations.
3) Explain following term related to I/O operation:
(i) Seek Time(ii)Rotational Delay (iii) Transmission Time

(B) Answer any one of the following: [4]

1)What is a Virus? Write any three infecting methods of Virus.
2) Define Binary Tree. Draw a tree diagram for following expression: E = [(a-b-c) +(a+ b -c)]$^{3/2}$

Q.4(A) **Answer any two of the following:** [6]

1) Explain time sharing related to process management of operating system.
2) Write C++ declaration for the following:
   i) Array of 15 integers
   ii) A pointer to an array of 10 doubles
   iii) Initialize an array with 5 float elements
3) Write an algorithm for Linear Search Method

(B) **Answer any one of the following:** [4]

1) Define Linked List. Draw and explain labeled diagram of linked list with 4 nodes.
2) Explain the following:
   (i) External Priority   iii)Purchase Priority
   (ii) Internal Priority   iv)Time Slice

Q.5 **Answer any two of the following:** [10]

(1) Write a program in C++ to accept 10 integers in an array and find its sum and average.
(2) Write a C++ function to find surface area of a sphere. Where A= 4 pi r^2
(3) Write HTML code for web page displaying the following table:

City	Wind speed	From	To
Mumbai	39	South	West
Raigad	47	South	West
Panaji	42	East	South

OR

Q.5 **Answer any two of the following:** [10]

(1) Write a program in C++ to read a set of 10 numbers from keyboard and find out the largest number in the given array.
(2) Write a Class-based C++ program to accept two integers and find its G.C.D (Greatest common divisor)
(3) Write the exact output of the following HTML code:

```
<HTML>
 < BODY>
        <TABLE border ="3" cellspacing ="10" >
        <TR>
                <TH Colspan ="3" > STREAM </TH>
        </TR>

        <TR>
                <TD> <A href =" science.html"> SCIENCE   </A> </TD>

                <TD > <A href ="  commerce.html"> COMMERCE   </A> </TD>

                <TD> <A href =" Arts.html">   ARTS     </A> </TD>
  </TR>
   </TABLE>
</BODY >
</HTML>
```

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ACA/DI/22	Prelim 1 Examination	Academic Year : 2021-22
Rev : 00	Subject : Computer Science2	Total Marks : 50
Class : 12th Science		Date : 04/01/22
Time : 9 .00 am to 12. 00 noon		

Instructions:

Draw Diagram wherever necessary . All questions are compulsory . Assume suitable data wherever necessary

2-83 12th Com.Sci II 1/2

Q. 1 A) Select the correct alternative and rewrite the following statements. **(4)**

1) In 8085 microprocessor, serial data from external device is received on .... Pin.
   a) SID b) SOD
   c) HOLD d) READY
2) In 8085 microprocessor, flag register is not affected after the execution of .... instruction.
   a) INR r b) DCR r
   c) ADD r d) INX rp
3) .... is not a characteristic feature of 8051 micro controller
   i) 4 kbyte of internal RAM b) 4 byte of internal ROM
   iii) 4 parallel bidirectional I/O port d) full featured serial port
4) A device used for modulation and demodulation process in network is
   a) Hub b) Router
   c) modem d) Repeater

B) Answer any two of the following. **(6)**

1) Draw neat labelled block diagram of a generic microprocessor.
2) Explain register indirect and immediate addressing modes in case of 8085 microprocessor with the help of suitable examples.
3) Mention any six important features of micro controller 8051.

Q.2 A) Asnwer any two of the following. **(6)**

1) Explain the function of the following registers of 8085 microprocessor.
   i) Program counter ii) Stack pointer iii) H.L. register
2) Explain following instructions of 8085 microprocesor with suitable example of each
   i) CMA ii) RAR iii) XRA A
3) Explain in short the six important characteristics of transmission media.

B) Answer any one of the following. **(4)**

1) Write a short note on evaluation of microprocessors. Give one example of each generation.
2) List the advance microprocessors of INTEL X-86 family and mention three attributes of anyone of them.

[PTO]

Q.3 A) **Answer any two of the following.** **(6)**

1) List any three primary functions of the CPU of microprocessor
2) Describe in brief the function of following pin in 8085 microprocessor
   i) HLDA ii) READY iii) RST 7.5
3) Explain the ring topology and token passing.

B) **Answer any one of the following.** **(4)**

1) Write a short note on flag register of 8085 microprocessor Explain the significance of flag bits with one example.
2) Explain in detail how 8051 microcontroller addresses two memory spaces.

Q.4 A) **Answer any two of the following.** **(6)**

1) The Accumulator of 8085 contains data 43H. What will be it's contents of the execution of following instructions independently.
   i) CMA ii) ANT 09H iii) INRA
2) Explain the following instructions with suitable examples
   i) DAA ii) LHLD
3) What is meant by protocol ? Explain the concept of TCP/IP protocol.

B) **Answer any one of the following.** **(4)**

1) State four LAN wireless transmission methods explain any two of them.
2) Write the functions of each of the following devices in short.
   i) Modem ii) Hub iii) Repeater iv) Router

Q.5 **Answer any two of the following.** **(10)**

1) Write an assembly language program to copy a block of data having starting address 8900H to the new location starting from 9100H. The length of block is stored at memory location 88FF H.
2) Write an assembly language program to add two 8 bit BCD numbers stored at memory location address 4100H and 4101H store the result at memory location 4102H onwards starting with least significant bit.
3) Write an assembly language program to find out Z's compliment of five numbers stored from memory location 3330H and onward store the result from memory location address 4100H.

OR

Q.5 **Answer any two of the following.** 10

1) A block of data is stored in memory location from 9101 H to 91FF H write an assembly language program to transfer the block in reverse order to memory location 9200 H and onwards.
2) Write an assembly language program to count the number of odd data bytes occuring in a block starting from the memory location address 7501H to 756FF H store the result at the memory location 7600H.
3) Write an assembly language program to perform multiplication to two 8-bit numbers where multiplicard is stored at the memory location 2501 and 2502H and multiplier is stored at 2503H. The result is to be stored. at memory location 2504H and 2505H.
   (Note : 8 bit multiplic and is extended to 16 bit)